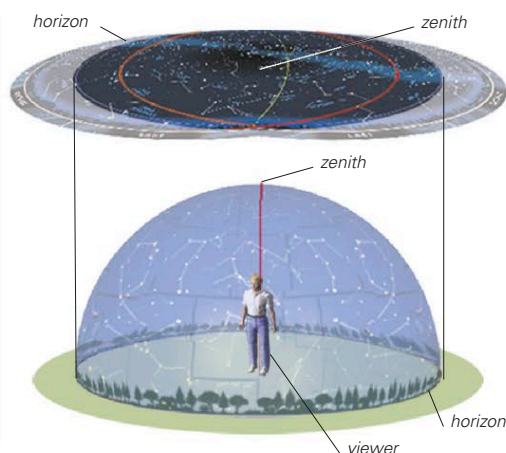


THE WHOLE-SKY CHARTS

The introduction to each month is followed by two whole-sky charts. These show the position of the stars at 10:00 p.m. local time on the 15th day of the month for both the Northern and Southern Hemispheres. They project the half of the celestial sphere (see pp. 62–63) that would be visible to a viewer under perfect conditions—that is, without any obstruction to the horizon. Any given star rises 4 minutes earlier each night compared to the previous night. Thus, the night sky changes subtly from one night to the next and even more dramatically from one month to another. To use the whole-sky charts, determine the color-coded horizon and zenith for your location (below), turn to the appropriate month, and position yourself and the whole-sky chart (right).

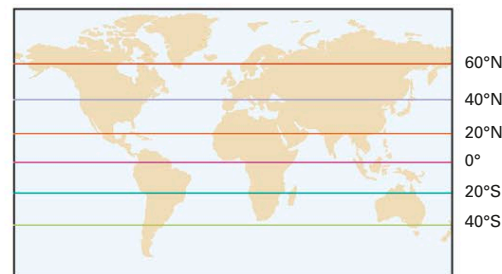
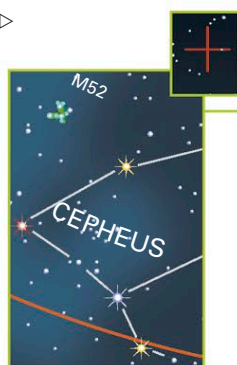


△ CELESTIAL SPHERE

Each whole-sky chart shows an area that equals more than half a celestial sphere because it combines three different projections of the night sky, as seen from three different latitudes on Earth. Each month, the sky charts show the night sky as it appears from 60°–20°N on the Northern Hemisphere chart and from 0°–40°S on the Southern Hemisphere chart.

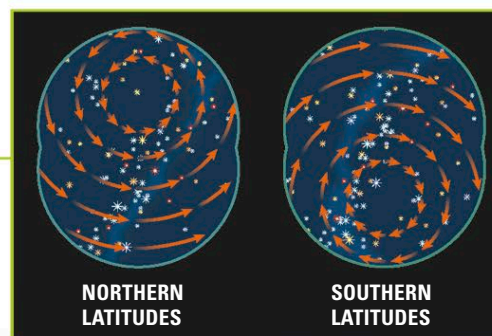
HORIZONS AND ZENITHS ▷

The stars located near the center of each chart can be seen on the zenith (the point directly overhead), while the stars near the chart's edge appear close to the horizon. Color-coded lines and crosses are used to identify the horizon and zenith on each of the three latitude projections on each monthly chart.



△ LINES OF LATITUDES

Determine the latitude line that is closest to your geographical location and use the color-coding on the sky charts to find the view from your location. Note that a 10° difference in latitude has little effect on the stars that can be seen.



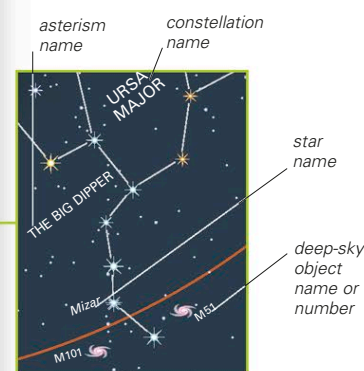
▷ STAR-MOTION DIAGRAMS

These diagrams show the direction in which the stars appear to move as the night progresses. Stars near the equator appear to move from east to west, while circumpolar stars circle around the celestial poles without setting.



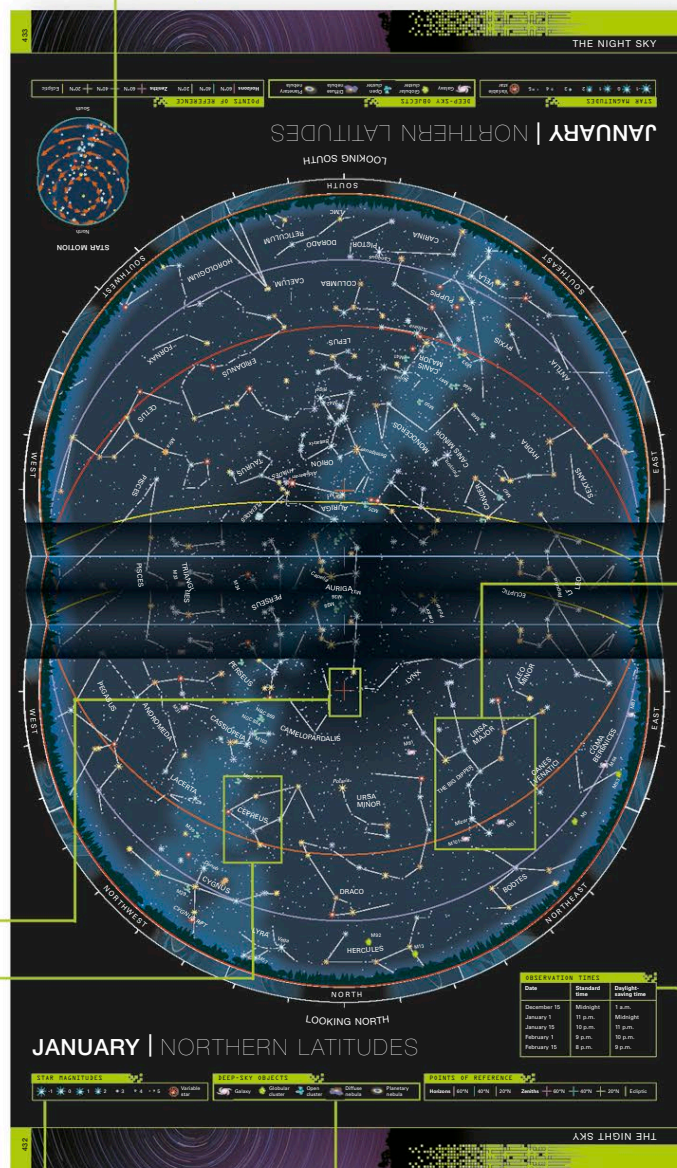
△ ORIENTATION

To view the sky to the north, face north and hold the map flat, with the NORTH label closest to your body. One of the color-coded lines around the near edge of the map will relate to the horizon in front of you. To view the south, turn around and reposition the map.



△ MAIN FEATURES

All 88 constellations are featured on the whole-sky charts, as well as any notable deep-sky objects within their boundaries. Well-known and easily recognizable stars, star clusters, and asterism patterns (see p.72) are also labeled.



OBSERVATION TIMES

Date	Standard time	Daylight saving time
December 15	Midnight	1 a.m.
January 1	11 p.m.	Midnight
January 15	10 p.m.	11 p.m.
February 1	9 p.m.	10 p.m.
February 15	8 p.m.	9 p.m.

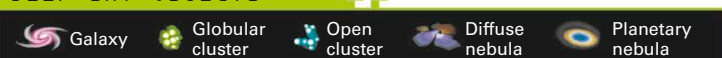
△ OBSERVING TIMES

Each chart shows the sky as it appears at 10:00 p.m. local standard time, mid-month. However, this view can also be seen at other times of the month, as well as 1 hour later, when local daylight saving time is in use. To view the sky at a time before or after 10:00 p.m., you may need to consult a different monthly chart.

▽ DEEP-SKY OBJECTS

Icons are used to represent a selection of deep-sky objects of interest to the amateur astronomer.

DEEP-SKY OBJECTS



▽ STAR MAGNITUDE

Stars that appear brighter than magnitude 6 are illustrated on the whole-sky charts. This key can be used to gauge their magnitude. About 25 prominent stars are also labeled with their popular names.

STAR MAGNITUDES

